

FIN305 Risk Management for Business

Week 5-6 Questions

Practice questions and problems

The numbers in parentheses after the questions correspond to the original numbering used in the textbook, which is intended to assist students in easily finding the relevant questions within the book.

1. The volatility of an asset is 25% per annum. What is the standard deviation of the percentage price change in one trading day? Assuming a normal distribution with zero mean, estimate 95% confidence limits for the percentage price change in one day. (8.2)
2. Assume that an index at close of trading yesterday was 1,040 and the daily volatility of the index was estimated as 1% per day at that time. The parameters in a GARCH(1,1) model are $\omega = 0.000002$, $\alpha = 0.06$, and $\beta = 0.92$. If the level of the index at close of trading today is 1,060, what is the new volatility estimate? (8.11)
3. A company uses the GARCH(1,1) model for updating volatility. The three parameters are ω , α , and β . Describe the impact of making a small increase in each of the parameters while keeping the others fixed. (8.13)
4. The parameters of a GARCH(1,1) model are estimated as $\omega = 0.000004$, $\alpha = 0.05$, and $\beta = 0.92$. What is the long-run average volatility and what is the equation describing the way that the variance rate reverts to its long-run average? If the current volatility is 20% per year, what is the expected volatility in 20 days? (8.14)
5. Suppose that GARCH(1,1) parameters have been estimated as $\omega = 0.000003$, $\alpha = 0.04$, and $\beta = 0.94$. The current daily volatility is estimated to be 1%. Estimate the daily volatility in 30 days. (8.16)

6. Suppose that the price of an asset at close of trading yesterday was \$300 and its volatility was estimated as 1.3% per day. The price at the close of trading today is \$298. Update the volatility estimate using the GARCH(1,1) model with $\omega = 0.000002$, $\alpha = 0.04$, and $\beta = 0.94$. (Hint: please consider the simple/arithmetic daily return.) (8.19)
7. The probability that the loss from a portfolio will be greater than \$10 million in one month is estimated to be 5%. What is the one-month 99% value at risk (VaR) assuming the change in value of the portfolio is normally distributed with zero mean? (8.23)

Solutions

Each week, students should attempt the provided exercise questions after lectures. Solutions can be found at the end of the textbook to check your answers. For those questions to which no solutions can be found in the textbook, the detailed solutions/explanations will be shown in the present file as follows. Selected questions will be discussed in tutorials, where the instructor will offer more detailed solutions. Recordings of the related tutorials will be made available to all students via Learning Mall.

Question (6)

- The proportional change in the price of the asset is $-2/300 = -0.00667$.
- Using GARCH(1,1), the conditional variance is updated as follows:

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2 = 0.000002 + 0.04 \times (0.013)^2 + 0.94 \times (0.00667)^2 = 0.00016264$$

Thus, the new daily volatility is $\sqrt{0.00016264} = 0.01275$, or 1.275% per day.

Question (7)

- The 99% VaR is $10 \times N^{-1}(0.99)/N^{-1}(0.95) = 14.14$ or \$14.14 million.